

Unit-II

Lecture: 3

Implementation of Linear Search & Binary Search in Python

Implementing Linear Search in Python:

```
def search(arr, n, x):  
  
    for i in range(0, n):  
        if (arr[i] == x):  
            return i  
    return -1  
  
# Driver Code  
arr = [23, 30, 4, 10, 40]  
x = 10  
n = len(arr)  
  
# Function call  
result = search(arr, n, x)  
if(result == -1):  
    print("Element is not present in array")  
else:  
    print("Element is present at index", result)
```

Implementing Binary Search in Python:

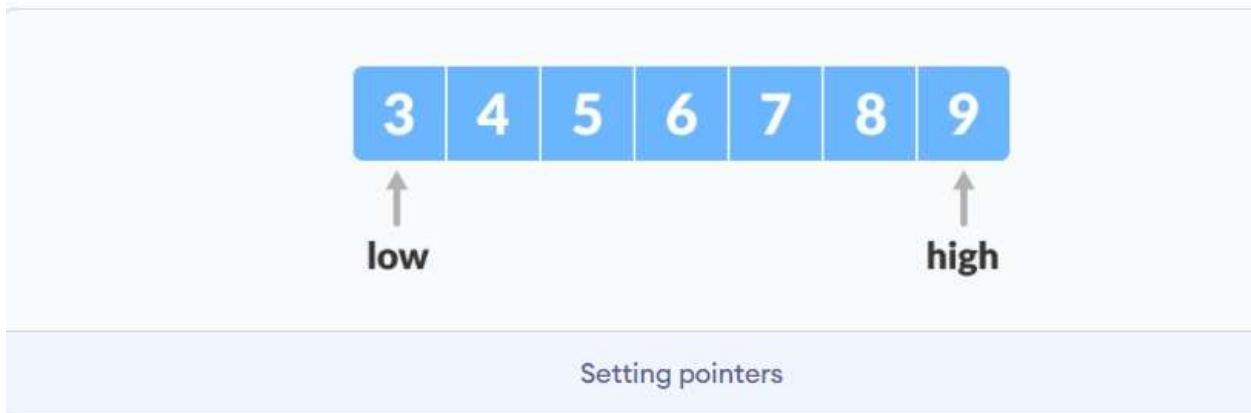
Binary search algorithm can be implemented in two ways:

- Iterative Method
- Recursive Method



Let $x=4$ i.e. 4 is to be searched in the above given array.

So, first we will set two pointers low and high at the lowest and the highest positions respectively.



Now, we have to find the middle element($\text{mid} = \text{low} + (\text{high} - \text{low}) / 2$)



Now, if $x == \text{mid}$, then return mid. Else, compared the element to be searched with m.

If $x > \text{mid}$, compare x with the middle element of the elements on the right side of me $\rightarrow$ we have to set low to $\text{low} = \text{mid} + 1$.

else, compare x with the middle element of the elements on the left side of mid $\rightarrow$ we have to set high to $\text{high} = \text{mid} - 1$.

We have to repeat this until low meets high, thus either x is found or not found.

Iterative Method:

Binary Search in python

def binarySearch(array, x, low, high):

*# Repeat until the pointers low and high meet each other
while low <= high:*

mid = low + (high - low)//2

*if array[mid] == x:
 return mid*

*elif array[mid] < x:
 low = mid + 1*

*else:
 high = mid - 1*

return -1

*array = [3, 4, 5, 6, 7, 8, 9]
x = 4*

result = binarySearch(array, x, 0, len(array)-1)

*if result != -1:
 print("Element is present at index ", result)
else:
 print("Not found")*

Recursive Method:

Binary Search in python

def binarySearch(array, x, low, high):

if high >= low:

mid = low + (high - low)//2

If found at mid, then return it

if array[mid] == x:

return mid

Search the left half

elif array[mid] > x:

return binarySearch(array, x, low, mid-1)

Search the right half

else:

return binarySearch(array, x, mid + 1, high)

else:

return -1

array = [3, 4, 5, 6, 7, 8, 9]

x = 4

result = binarySearch(array, x, 0, len(array)-1)

if result != -1:

print("Element is present at index " + str(result))

else:

print("Not found")